Research Project

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Degree: Honors (Information Systems and Technology)

Title of Thesis:

Factors that influence the adoption of Adaptive Learning Systems in higher education.

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Abstract

Over the past few years, higher education institutions have integrated the use of computers and online learning, both on and off campus. Adaptive learning is a type of learning that involves the use of computer algorithms and computers to facilitate and deliver customised learning materials based on student preferences. Adaptive Learning can play a major role in improving student academic performance. When the machine learning algorithms are integrated in Learning Management Systems (LMS), they can make it easier for teachers to provide materials to students in a customised way, making it easier for students to understand these academic materials. Adaptive learning technologies are used to create learning environments that are student-oriented. There have been few researches that have explore the advantages of having an adaptive learning system and they showed that indeed adaptive learning can improve student's overall performance. There hasn't been a lot of implementation of these algorithms in LMS and this study is aimed at examining the factors that influence the adoption of Adaptive Learning Systems in higher education in South Africa. The research will use a case study approach comparing data collected from two private higher education institutes namely Pearson Institute of Higher Education and IIE Monash South Africa.

Keywords**:** Algorithm; students; machine learning; adaptive technology; learning management system; artificial intelligence; SCORM; Moodle; Distance learning, learning styles, design methodology; Learner Model; Adaptive learning systems; Felder-Silverman Learning Style Model; Web usage mining techniques; machine learning algorithms; K-modes clustering algorithm;

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Chapter 1: Introduction

The introduction of the use of internet and its technologies in the education sector, has transformed the way the process of teaching and learning is done. These internet technologies are being used intensively in improving students' performance. Now there are different ways in which teachers can teach and how students can learn. This study will explore the factors that influence the adoption of Adaptive Learning Systems in higher education in South Africa.

Adaptive learning is a type of learning that involves the use of computer algorithms and computers to facilitate and deliver customised learning experience based on student's characteristics (Tseng *et al.*, 2008). It involves the use of machine learning algorithms in learning management systems (LMS) and computers to alter the presentation of learning materials based on student preferences (Tseng *et al.*, 2008). Its main aim is to create a customised learning experience for each student, in connection with his or her unique needs. It is able to provide pathways, just-in-time feedback and resources. Adaptive learning systems provides a unique experience rather than providing a learning experience that is one-size-fits-all.

1.1 Background of the study

The study is aimed at examining the factors that influence the adoption of Adaptive Learning Systems in higher education in South Africa. Over the past few years, higher education institutions have integrated the use of computers and online learning, both on and off campus. The teachers use the online environment to provide feedback and assessments to students (Sawacki-Richter and Latchem, 2018). The use of IT technologies has allowed for collaborative online learning. Academics and students use digital media, laptops, and mobile devices to work together, eliminating the geographical barrier.

Adaptive Learning can play a major role in improving student academic performance. The adoption of Adaptive Learning systems can make it easier for teachers to provide materials to students in a customised way, making it easier for students to understand these academic materials (Ueda *et al.*, 2018). Adaptive learning technologies are used to

create learning environments that are student-oriented (Kara and Sevim, 2013). Through the use of Adaptive learning technologies, students can be classified into groups according to the their learning styles (AISSAOUI *et al.*, 2019). This makes it easy for academics to understand their students and help them accordingly to improve their academic performance. This is way it is important to adopt these machine-learning systems.

1.2 Problem Statement

Internet and IT technologies have indeed smooth lined the teaching and learning process (Assar, 2015). The use of technology has profoundly changed the way education is delivered and conceived. When ICT is integrated and used well in the educational process, it can significantly enhance performance of students (Assar, 2015). The perception students have towards IT technology can also affect how learning goal are achieved. If students perceive the use of ICT as easy and useful to them, then this can indeed improve their performance. Introduction of adaptive learning systems in education has opened new possibilities on how technology can help facilitate the process of learning and teaching. Through the use of adaptive learning systems, students' learning preferences can be considered during development of study materials and this can improve their academic performance (Assar, 2015).

The aim of the study is to find out what factors influences the adoption of adaptive learning systems in higher education in South Africa. Many researches on adaptive learning systems show that when learning resources are personalised they increase learner's comprehension, interest and increase the learning success (El Guabassi et al., 2018).

Thus, it is important to consider using adaptive-learning technologies in higher education in South Africa. The reality is that, there has been less or no interest in implementing these machine learning algorithms in most higher education institutes in the country (El Guabassi *et al.*, 2018). The reason for this is that the adoption of these technologies may include lack of resource accessibility and availability, difficulties in implementing tools, and the difficulties in identifying learning styles (El Guabassi *et al.*, 2018).

1.3 Research Questions

- Which adaptive technologies can be used in higher education?

- What are the requirements for using adaptive learning systems in higher education?
- What are the advantages of using adaptive learning systems in academics?
- What are the people factors that influence the use of Adaptive learning systems in higher education?

1.4 Research Objectives

- To examine the current adaptive technology usage in higher education in order to improve students' performance.
- To determine the requirements for using adaptive technology in higher education to improve students' performance.
- To determine the advantages of using adaptive learning systems in higher education to assess the usefulness.
- To examine the people factors that influence the use of Adaptive learning systems in higher education in order to supplement traditional (face-to-face) teaching style.

1.5 Significance of the study

The study will help to supplement both teaching and learning processes in higher education through the implementation of Adaptive learning tools in higher education. This will help academics to offer more help to the students and make students more interested in the learning material while making the material easy to understand (Tseng *et al.*, 2008). The results in the study will act as a foundation where more research could be built upon. The research will also provide several ways in which the adaptive technologies can be implemented for more practical research projects in higher education.

1.6 Summary

This chapter introduces the research topic. The researcher would like to investigate the factors that influence the use of adaptive learning management system in education. It is clear from this chapter that the use of adaptive learning systems can enhance learning process. The researcher goes on to describes the research problem statement. The research questions are also identified that will help to answer the research topic.

Chapter 2: Literature Review

2.1 Introduction

This chapter will discuss some of the work that has been done under adaptive learning systems. It will first look at the adaptive machine learning and then go ahead to discuss some of the common uses of adaptive machine learning systems. This chapter will also look at the history of adaptive learning management systems and how it has been used in education. In addition, different learning styles and learning challenges faced by students will be explored. This chapter will show how these learning challenges and learning styles are used in making a learning management system adaptive. Last but not least, this chapter will look at few case studies on the implementation of adaptive learning management systems in school.

2.2 Adaptive machine learning

Adaptive machine learning is a tool that uses artificial intelligence (AI) through the use of statistical techniques to create intelligent computer systems that can learn from databases (AISSAOUI *et al.*, 2019). Artificial Intelligent is defined as the ability of a computer controlled robot or digital computer to perform different tasks normally performed by intelligent beings, in this case, human beings (Wang *et al.*, 2019). The tasks can include the ability to discover meaning, reason, learn from the past experience and generalise things.

Adaptive machine learning makes use of different algorithms. An algorithm can be defined as a set of rules or a process that are performed using a computer in calculations or in problem-solving operations (Crassidis and Markley, 1997). Algorithms are used for different processes. Some uses include, performing automated reasoning, calculation and data processing.

2.2 Common uses of Adaptive machine learning

2.2.1 Image Recognition

Machine learning is used vastly in image recognition. For example, a black and white image can be recognised by using the intensity of each color pixel (Kaur, 2019). A colored image contains 3 types of pixel which are red, green and blue (RGB). So, if a picture has lower intensity in these pixels, then the computer can recognise that the image is black and white. A black and white color image has low intensity of each pixel (Kaur, 2019).

Machine learning is also used in face detection in an image. If an image exists in the database, then the system is able to compare the pre exiting picture in the database with the new image it is trying to dictated. This method can also be used to discern printed letters and handwritten by recognition the characters that are used in the document (Kaur, 2019).

2.2.2 Speech Recognition

Speech recognition is the process of translating the spoken words into text. It is also referred to as the automatic speech recognition or computer speech recognition (Kaur, 2019). Machine learning tools are used to recognise the spoken words from a clip file or audio, and then the specific spoken words are transformed form audio into a text-based file (Kaur, 2019).

We see the application of these machine learning tools in most of the voice user interfaces. Examples include data entry for documentation, voice dialing, appliance control, and call routing (Kaur, 2019).

2.2.3 Medical diagnosis

There has been a vast increase in the use of machine learning tools in the healthcare sector (Kaur, 2019). Machine learning tools are used to diagnosis different types of diseases. The tools make prognosis by analysing the clinical parameters and their combinations. These prognoses are used for patient monitoring, therapy planning and even for further medical research about the particular disease (Kaur, 2019).

2.2.4 Learning associations

Learning associations can be defined as the process that involves developing a deep insight of various associations between products (Kaur, 2019). A simple example will be the case where unrelated products are associated with one another. Machine learning is used in most shopping website where the system is able to show different associations between products that people usually buy. When a customer buys one product, the customer is usually shown other products that other buyers have bought together with the current selected item. Other products that have- similar and same usability with the selected product can also be displayed. These machine learning tools are also used in marketing strategies in the case where a new product is being launched into the market. The association between the old product and the new product is created in order to boost sales of the new product (Kaur, 2019).

2.2.5 Classification

Machine learning is used in classification processes. classification can be defined as the process of putting each item or individuals in a class based on similar characteristics (Kaur, 2019). Analysts use data in order to explore efficient relationships between items. For example, in the banking sector, before a person is granted a loan, machine learning tools are used to assess the customer if they can be able to pay back the loan. The system looks at several characteristics like savings, earnings, and financial history to determine the customer's eligibility to access the load. At the end of the day, the system reveals if the customer should be granted the loan or not (Kaur, 2019).

2.2.6 Extraction

Another area that machine learning is used intensively is in data extraction. Information extraction is the process of extracting relevant information from data that is unstructured (Kaur, 2019). Machine learning tools are used in big data and larger data warehouse. Big data and Data warehouse are databases that store large amounts of data (Kaur, 2019). The data is collection from different sources like the web pages, business reports, blogs, articles and emails. These databases use machine learning tools to transform these huge data into meaningful information (Kaur, 2019).

2.3 Adaptive machine learning in education

2.3.1 Learning management system (LMS)

A learning management system (LMS) is a system that is used to facilitate the teaching and learning process by managing all the required learning resources (Webdesign, 2019). It makes it easy for student to access materials like lecture slides, unit guides and as a reading resource. It is also used for submission of assessments and as a communication platform in higher education.

The first LMS was developed in the year 1924 by Sidney Pressey (Webdesign, 2019). The LMS was initially called the Teaching Machine (Webdesign, 2019). After this invention, there were several improvements built upon the first teaching machine. In the year 1956, Gordon Park and Robin McKinnon-Wood invented the first Adaptive teaching system (Webdesign, 2019). The Adaptive teaching system was called SAKI (Webdesign, 2019). The system was able to automatically adjust the difficulty levels of questions according to performers' level of learners (Webdesign, 2019).

2.3.2 Learning challenges

Many factors can contribute to the success and the failure of students. The use of technology has brought and helps the academic world to utilise technology in facilitating learning process in order to reduce the failure rate and increase student's engagement.

A research conducted in United States of America on selected academic institutions (Fook and Sidhu, 2015), identified several challenges faced by instructors, postgraduate students and undergraduate students in selected education institution. The research identified eight main challenges faced by the leaners and they included becoming an active learner, following instructional problem, cognitive challenge, coping with reading materials, burden of assignments, language barrier, culture difference in higher education and time management. The research concluded that there was a need for academic institutions to put more emphasis on developing knowledge through student and academics' interactions. To be able to do this, there was a need use adaptive learning

technologies in order to develop appropriate learning and teaching resources that could reduce the challenges.

Adaptive learning systems are implemented using adaptive machine learning to adapt learning paths, presentation styles and learning content based on individual students' learning status , profiles, and human factors (Hwang et al., 2013). Through the implementation of adaptive learning systems, most of these challenges can be reduced and the performance of the students can be improved.

2.3.3 Algorithms and their uses

There are different ways that an LMS can have adaptive machine learning capabilities. Like mentioned early, the learning machines use algorithms that help a system act like an artificial intelligent system. Adaptive learning is a type of learning that involves the use of computer algorithms and computers to facilitate and deliver customised learning materials based on student preferences (Tseng *et al.*, 2008). These computer algorithms can be used in online tests and class exercises, and the algorithms can also be included in platforms like Moodle (Hubalovsky, Hubalovska and Musilek, 2019). Over the years, there has been a number of algorithms that have been developed and can be integrated into leaning management systems to facilitate leaning and teaching processes in school. Below are different ways algorithms are used in LMS:

- **Curriculum Sequencing algorithm:** This algorithm is used to provides each student with the most suitable sequence to be applied when leaning topics in the subject or unit they are enrolled in (Hwang et al., 2013). It rearranges the study content in such a way that it is easy for the particular student to understand the material.

- **Intelligent Analysis algorithm:** This algorithm is used to identifies students' solutions on test and quiz, then it provides learning supports according the solutions (Hwang et al., 2013). If a student's solution indicates that the student does not know anything about the question, the system will provide all the necessary material (from basics) to help the student understand the learning

materials. In the case where the student has some knowledge about the study material, then the system will also provide the knowledge that seem to be missing from the student's solution. In this case every student is helped according to their unique needs.

- **Example-based Problem-Solving algorithm:** The example based algorithm provides different suggests of examples or cases to students to help them solve the problem at hand (Hwang et al., 2013). The system is able to provide specific examples on how a particular question can be solved and this acts as a template on which the students can follow and be able to solve the problem. This also provides more insights on the problem at hand as students can be able to understand the question more based on the example provided by the system.

- **Adaptive Presentation algorithm:** This algorithm is used to provide customised learning experience to each student by adapting the learning content based on the student's personal characteristics or the knowledge level of the student in the subject area (Hwang et al., 2013).

- **Adaptive Navigation Support algorithm:** This algorithm is used to provide students with personalised learning paths or links based on their personal characteristics or knowledge levels (Hwang et al., 2013). For example, if the system detects that the student has low levels of understanding of the subject, the system will provide links to the students that will enable the student to understand the learning materials from the basic level. On the other hand, if the system detects that the student has a good understand of the material, then the system will provide links to more advanced materials to enhance the student's knowledge and understanding.

2.4 Implementation of adaptive learning tools in education

There has been different implementation of adaptive machine learning tools in the different academic institutions and they have proven to be effective and improved

students' performance (Hwang *et al.*, 2013). The researcher will now discuss some of the examples where the adaptive learning machines were used to supplement the traditional teaching and learning processes.

2.4.1 Implementation of Adaptive presentation LMS

The paper titled "SCORMAdaptiveQuiz: Implementation of Adaptive e-Learning for Moodle", explains how a plugin (a software containing algorithm) was developed and how it was integrated into a learning management system (LMS) to be able to make the LMS adaptive (Ueda *et al.*, 2018). The LMS was integrated with a plugin that contained algorithms that changed the presentation of study material based on leaners knowledge level. The paper went on to implement and analysed the performance of the adaptive system in the year 2016. The data collected in the academic year of 2016 was measured against the previous data of the academic year of 2015 in order to measure the improvement of student's performance (Ueda *et al.*, 2018).

The machine learning system used pre-exiting knowledge of students about the level of understanding of different subjects. This data was found on online feedback platforms where the student responded if they enjoyed the study materials. Some students said the study materials were boring because they knew most of the things and some found it challenging because they did not have any prior knowledge of the study material.

A pre-test was conducted in order to collect different general levels of understand or determine the pre knowledge levels of student different course.

The plugin module included in Moodle was called SCORMAdaptiveQuiz. The plugin was used to perform three distinctive functions:

- **Setting function:** for determining different category of questions
- **Display function:** for providing the pre-test results in a table format
- **Learning state change function:** for assigning the learning status of student from the SCORMAdaptiveQuiz based on their score from the questions in the pre-test

After the implementation of the adaptive system, data was collected between 2015 to 2016 of the academic year. The data revealed two main improvements:

- **Increase in the students' performance:** it revealed a tremendous increase in the performance of the students. The final scores of students improved after the SCORMAdaptiveQuiz plugin was integrated in Moodle. The students performed better in the academic year of 2016 compared to the academic year of 2015.

- **Improvement in the access rate of the learning materials by student:** Incorporating adaptive learning tools in Moodle also increased the access and interaction of student with the learning materials on Moodle. Now the students become interested in the learning material as it was easy to understand them.

2.4.2 Implementation of Adaptive LMS to students' learning styles

A learning style is the combination of characteristic factors which include affective, cognitive, and physiological that are used to indicate how a leaner responds, interacts with, and perceives the learning environment (May Truong, 2015). This is directly related to how a student prefers to learn.

A machine learning tool was created to be able to automatically detect the learning style of the students through using machine learning algorithms and existing behavior data of learners (AISSAOUI *et al.*, 2019). A web that utilises mining techniques was used to pre-process the extracted data from the learning management system. Mining techniques are methods that are used to extract and process data to make meaning out of it (Kaur, 2019). The extracted data was pre-processed and learner's sequences were captured. The captured data was used as inputs in the K-modes clustering algorithm to categories the data into 16 different combination of learning styles. The K-modes of clustering is an algorithm that helps in grouping things into different groups or into k-clusters based on similarity in characteristics (AISSAOUI *et al.*, 2019).

During the learning process, student follow different learning paths and some of the student's characteristics are related to their learning paths (Van Seters *et al.*, 2012). At

the student level, there are different learning styles that can be identified. These might include: Convergent style, Accommodation style, Divergent style, and Assimilation style (Magdalena, 2015). In addition, the relationship between learning behavior, learning style and academic performance of the students can be established at this level. There are many theories on learning styles and one of the widely used is Felder–Silverman learning style (Hwang et al., 2013).

The learning styles where based on the Felder and Silverman learning style model (FSLSM). The Felder and Silverman Learning style Model it's a learning styles model that is use learning styles to classify learners accordingly (AISSAOUI *et al.*, 2019). The FSLSM utilise the concept of dimensions. It has four dimensions, with each dimension having two contrasting characteristics. These four dimensions are; input (visual/verbal), perception (sensing/intuitive), understanding (sequential/global), and processing (active/reflective) (AISSAOUI *et al.*, 2019). A learning style was defined through the combination of one category from each dimension which categorised the student into different types of learners. The students' list as outlined by AISSAOUI *et al.*,(2019).included:

- **Active learners:** These leaners understand and retain information well through a practical interaction with the learning material.

- **Reflective learners:** The leaners prefer to think and process the learning material quietly in form of reflection.

- **Sensing learner:** These leaners prefer real-world facts so that they can apply the learning material in a more practical sense.

- **Intuitive learners:** they prefer material that do not require routine calculations and memorisation.

- **The visual learners:** prefer the use of diagrams, pictures, and charts in order to understand the learning materials well. They like the use of visual representations in leaning.

- **The verbal learners:** they understand better when the learning materials are in the words format; both spoken and written.

- **Sequential learners**: they prefer learning in a linear way, going one chapter to another logically.

- **Global learners:** the leaners prefer learning in a randomised way. Jumping from one topic to another, without following the normal topic arrangement.

After the machine learning techniques were successfully implemented in the LMS, the system was tested and it showed that the LMS was able to predict different learning styles associated with each student. This was used to improve the student's performance as the system could provide the specific learning materials in the format that correlated with the specific learning style of the person (AISSAOUI *et al.*, 2019).

Adaptive learning plays a major role in making e-learning education successful. Adaptive learning systems can improve e-learning by providing a personalised learning path, or by presenting personalised content that suits the individual student (Hwang *et al.*, 2013). In many cases, most of the online or e-learning platforms lack the adaptively (Ueda *et al.*, 2018). Integrating these adaptive tools into the e-learning platform improved the learner's performance compared to when adaptively is not implemented in e-learning system (Hwang *et al.*, 2013).

2.5 Summary

The researcher looked at some of the literature under adaptive learning systems. The chapter also explored some of the learning challenges faced by student which includes, time management, burden of assignment and facing problems with following instructions, just to mention a few. The chapter went on to looked at some of the learning styles that

can be used to make learning management systems adaptive. Some of the learning styles include, active learner, intuitive leaner, global learner and verbal learner. From the case studies discussed in the chapter, it is clear that adaptive learning systems can improve academic performance of students.

Chapter 3: Research Methodology

3.1 Introduction

This chapter will look at the methodology that the researcher will use when collecting data that will be used to answer the research questions. The topics that will be discussed under this chapter include; research design, research approach, study site, target population and more. For the researcher to be able to answer the research questions, data that is collected must be analised and this chapter will also look at the limitations of the methodology.

3.2 Research Design

The research used a case study approach. This is a research method involved the exploration and interpretation of complex issues in the subject of study (Chaiklin, 2006). This will help to contribute to the scientific discipline of adaptive learning systems and allow further practical experiment to be built upon it. Case studies contribute to the systematic production of exemplars, which makes the discipline more effective. Without a large number of exemplars, a discipline becomes ineffective (Tight, Symonds and Symonds, 2016).

3.2 Research approaches/paradigm

The research will be conducted using quantitative approach (Anderson, 2010). Quantitative research involves steps such as collection of data, analysing of data, data interpretation, and documentation of study results (Leal Filho and Kovaleva, 2015). The researcher will use this approach so that the data that is collected from the sample size, can be analysed properly for conclusions to be drown from it.

3.3 Study Site

Pearson Institute of Higher Education (PIHE) and IIE Monash South Africa (IIE MSA) were the selected academic institutions where the research study was conducted. These two institutions were selected because they are most suitable to the study due to their academic facilities and geographical location.

3.4 Target Population

The target population for the study are students. This is because the research study is aimed at investigating the factors that influence the use of adaptive systems in education amongst students. Therefore, the population in the higher education institutions were students which is the appropriate target population for the study.

3.5 Sampling strategies

Sampling is very important as it helps to give a true representation of the targeted population through the use of a selected population size from the targeted population (Bollard and Browder, 2017).

The study used two most common approaches as the sampling strategies. The first sampling strategy that was used was purposeful sampling, which is also referred to as judgement sampling (Marshall, 1996). A framework or a criterion was used to asses which individuals are most likely to give productive answers to the question. The framework is usually developed by the researcher using available literature and practical knowledge of the research area (Marshall, 1996). In this case, the students at the academic institution were used to select the sample.

The second one is probability sampling, which is also called random sample. In a random sample, every member from the target population has the same chance of being selected to be part of the sample (Marshall, 1996). A stratified random sampling was used to allow smaller groups of students to be studied in more detail (Marshall, 1996).

3.6 Sample Size

In order to be able to make valid inferences of the target population, there is a need to select the optimum sample size. When a large sample size is used, the random sampling

error is reduced. In this case, people are likely to select a larger sample size. Although this night seems like a good idea, there is no real advantage of using a big sample size. This is so because the sampling error is inversely proportional to the square root of the selected sample size (Marshall, 1996). Therefore, for this study, the sample size that I selected was 100 students. Out of the 100, 50 students were from Pearson Institute of Higher Education and the remaining 50 students were from IIE Monash South Africa.

3.7 Data collection method

The data collection tools that was used were questionnaires. A questionnaire is a research tool that comprises of different questions used in the process of gathering information from selected respondents (Bryman, 2006). Structured questionnaire was used to collect data that helped to answer the research questions. Structured questionnaire comprised of a set of questions that were standardised and had a fixed scheme that was used to specify the order and the exact wording of the questions (Bryman, 2006).

Google Forms was used to administer the questionnaires. Google Forms is an application that is used for survey administration. It includes packages like Google Sheets, Google Drive office suite, Google Slides, and Google Docs (Google, 2018). This application was selected because it was easy to share and administrate the questionnaires as its features include sharing and collaboration (Google, 2018).

3.8 Data Quality Control

To make sure that the data that gathered during the research is reliable and valid, the trustworthiness was considered (Laumann, 2018). When conducting quantitative research, four aspects were taken into consideration and these aspects included: Credibility, Transferability, Dependability, and Confirmability.

Firstly, credibility looks at the confidence in the 'truth' of the findings. This helps to verify if the study has indeed measured what it is meant to measure (Connelly, 2016). Secondly, transferability shows whether the findings have any applicability in other contexts. Thirdly, dependability shows if the research findings could be repeated and are consistent. Lastly

confirmability shows the degree of neutrality of the findings. In other words, to what extent are the research findings shaped by the respondents and not the researcher's motivation, bias, and interest (Connelly, 2016).

3.9 Data analysis

The quantitative data that was obtained from the research was interpreted and organised in the following way: Inferential statistics was used to analyse the quantitative data. Inferential statistics is a statistical data analysis method that uses a random sample of data from the target population in order to make inferences about that target population (Bishop, Wonnacott and Wonnacott, 2007). Through this, a research conclusion was reached inferring the population view on the research study from the sample data (Trochim, 2006).

3.10 Ethical Consideration

In order to illuminate any ethical issues that might come up during the research, ethical clearance to conduct the study had to be approved by Pearson Institute of Higher Education. During the research, consent forms were handed to each respondent to make sure that their details are kept confidentially. A gatekeepers' letter was also obtained from IIE Monash South Africa tin order to collect data from the academic institution.

3.11 Limitation of the study

The first limitation of the study was that it only focused on the use of Adaptive learning technology in higher education at two private academic institutions. There are many areas in which adaptive technologies can be explored. These adaptive machine learning tools are used in the engineering field (Biamonte *et al.*, 2017), in business world (March, 1991) and in health sector (Barto, Sutton and Anderson, 1983) but because of time constraints, the study only focused on the usage and implementation of adaptive technologies in higher education.

Secondly, only two private higher education institutions namely, IIE Monash South Africa and Pearson Institute of Higher Education were considered in the research. Also, not all the educational tools were considered. This could limit the percentage of students that

were selected in the sample size if students are not willing to share their experience of adaptive technologies. This may have affected the results obtained in the research. In order to overcome these problems, questionnaires were conducted in all the departments of the selected academic institutions so as to reduce the possibility of having low response rate.

Thirdly, only the private academic institutions were considered in this study. More case study approaches could be used to compare the private and public schools on the use of adaptive technologies.

3.12 Summary

In this chapter, the researcher looked at the methodology that will be used to conduct the research and how it will be used to help to answer the research questions. The chapter clearly indicated the sample size which is 100 students, 50 from IIE MSA and 50 from PIHE. The researcher will use questionnaire as a data collection method. In order to make sure that the data collected is relevant to the research topic, trustworthiness will be use to access the reliability and validity of the data. The researcher also explained how ethical issues will be handles in case any issues arise during the data collection period.

Chapter 4: Findings and Data analysis

4.1 Introduction

This chapter will discuss the findings and analysis of the data that was collected through the use of questionnaires that was administered at two private higher education institutions: Pearson Institute of Higher Education (PIHE) and IIE Monash South Africa (IIE MSA). SPC (Statistical Process Control) for Excel is a statistical tool that was used to analyse the quantitative data. This chapter will provide insight on the quantitative analysis of the data collected. The data was analysed using different statistical tests and this includes:

- Descriptive statistics
- Kruskal-Wallis Test with Pairwise Comparison

- Chi Square Results for Goodness of Fit

4.2 Number of respondents

The total number of participants for the study was 100. This comprised of 50 students from IIE Monash South Africa (50%) and 50 students from Pearson Institute of Higher Education (50%). Figure 1 below shows the total number of students.

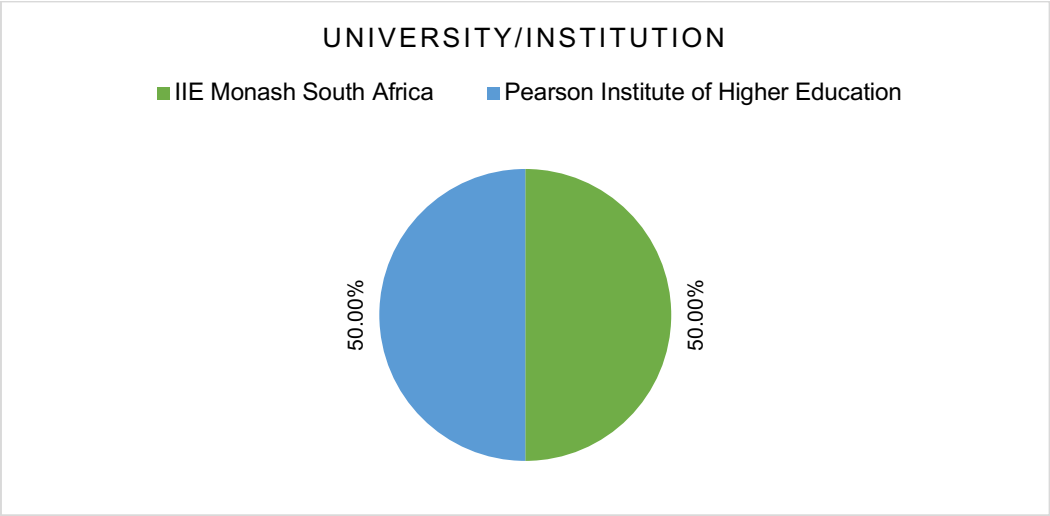


Figure 1. Number of students per institution

4.3 Learning environment factors

This section looks at the characteristics of the learning environment at the two institutions. This question was asked to assess the current teaching methods and facilities that are available at PIHE and IIE MSA. The themes under this section include:

- Teaching style
- Experience with learning facilities

4.3.1 Descriptive statistics of learning environment factors at IIE MSA

4.3.1.1 Teaching style

Teaching style is related to the different types of teaching methods, namely face-to-face, online or the mixed (online and face-to-face). A total of 70% of students at IIE MSA indicated that the teaching style is mixed and 24% of students indicated that their lessons

are in tradition face-to-face. The remaining 6% of the students indicated that their classes are completely online.

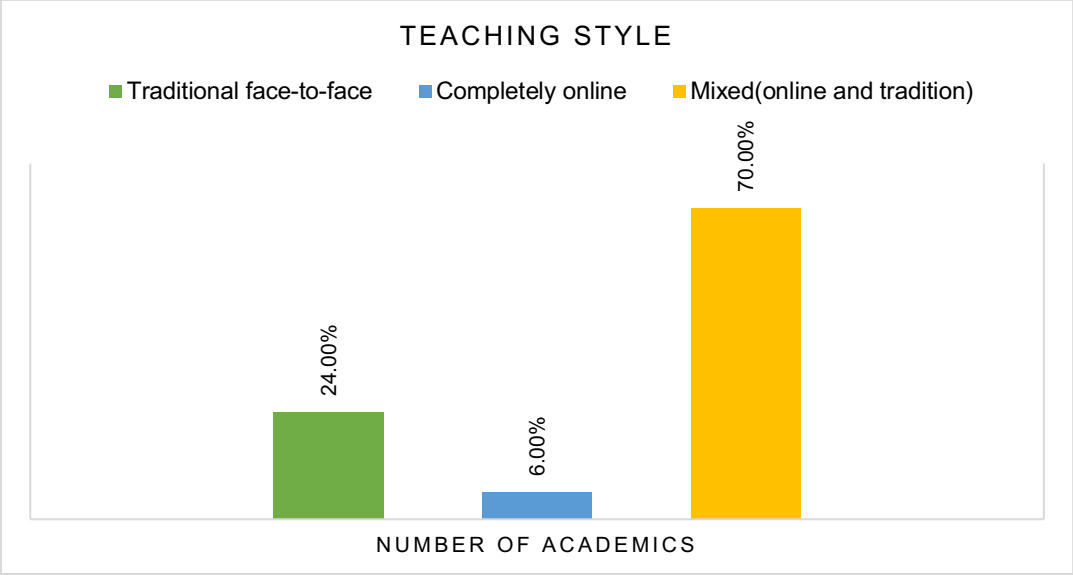


Figure 2. Teaching styles at IIE MSA

4.3.1.2 Experience with learning facilities

Experience with learning facilities is related to all the learning resources that are found at the education institution. In particularly, the resources that are used to facilitate the teaching and learning processes. The purpose of this question was to determine the current level of usage of these resources and how they help in the teaching and learning process.

4.3.1.2.1 E-Classroom

E-classrooms comprises of resources like computers, projectors, lecture capture systems and SMART boards that are used during teaching and learning process. The findings showed that 34% of the students indicated that the resource was poor. Meaning that, the facility is available but not in good condition to be utilised. The findings also showed that 14% of the students indicated that, the facility is fair (utilisable, but not in its best condition) and 8% indicated neutral as their response. Furthermore, 28% of the students indicated

that the facility is good and the remaining 16% indicated that the facility is excellent. From the data, we can see that the current level of usage of e-classroom is high at IIE MSA and it aids in the process of teaching and learning.

4.3.1.2.2 Learning Management System (LMS)

A Learning Management System (LMS), is a tool that is used to organise and share learning resources. At IIE MSA, 38% of the students indicated that the facility is available but not in good condition to be utilised, 12% indicated that LMS is utilisable but not in its best condition, 8% of the students indicated there response as neutral, 26% indicated that the facility is good and the remainder 16% of students indicated that LMS is excellent. The results show that, the Learning Management System is highly used in the process of teaching and learning at IIE MSA.

4.3.1.2.3 Plagiarism detection software

This resource is used to prevent students from presenting someone's work as their own. An example of this resource is Turnitin. From the data collected, only 2% of the students indicated that the resource is not in good condition to be utilised, 46% indicated that the software can be used, but it , 4% of the students were neutral, 28% of students indicated that Plagiarism detection software is in good condition and 20% indicated that the facility is in excellent state. The data clearly shows a high usage level of the facility in the process of teaching and learning. These findings are displayed in Figure 3 below.

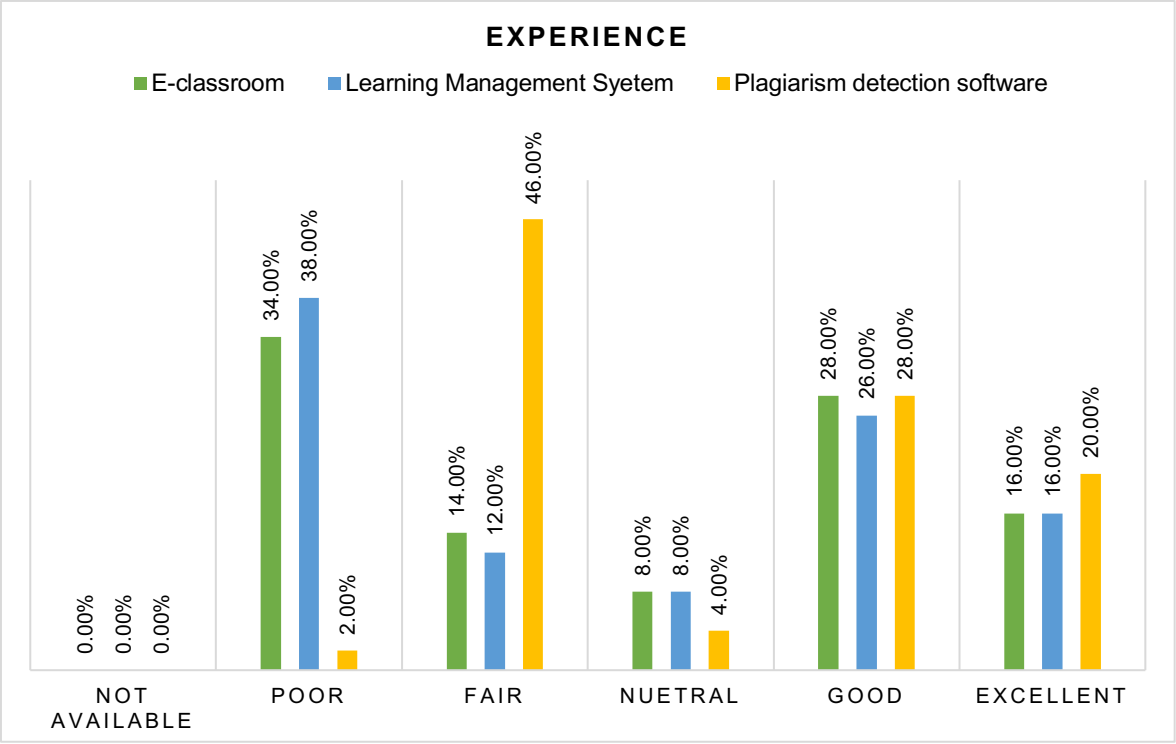


Figure 3. Experience with learning facilities for IIE MSA

4.3.2 Descriptive statistic of learning environment factors at PIHE

4.3.2.1 Teaching style

At PIHE, a total of 72% of the students indicated that the teaching style is mixed and the remaining 28% indicated that their lessons are in traditional face-to-face teaching style.

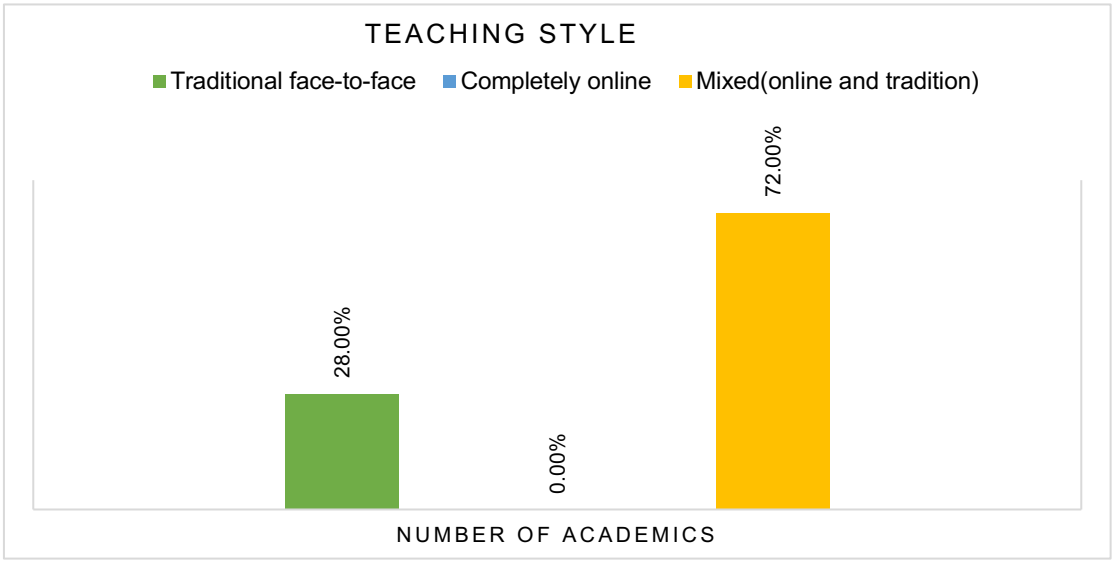


Figure 4. Teaching styles for PIHE

4.3.2.2 Experience with learning facilities

4.3.2.2.1 E-Classroom

Based on the data collected at PIHE, 24% of students indicated that the resource is available but not in good condition to be utilised, 26% indicated that the facility is utilisable, but not in its best condition, 20% indicated that it is neutral, 28% of the students indicated that the facility in good condition and only 2% of the students indicated that the facility is in excellent condition. From the data collected, it is evident that e-classroom is used extensively in the teaching and learning process at PIHE.

4.3.2.2.2 Learning Management System (LMS)

At PIHE, 30% of the students indicated that this facility is available but not in good condition to be utilsed, 12% indicated that LMS can be utilised but it is not in its best condition, 15% indicated that the facility is neutral, 25% indicated that it is good and the remailing 18% of the students indicated that the facility is excellent. It is evident from the data that, students use Learning Management System in their learning process.

4.3.2.2.3 Plagiarism detection software

Based on Figure 4.5 below, 3% of the students indicated that the resource is available but not in utilisable, 40% indicated that the facility is utilisable but not in it best condition, 10% indicated that it is neutral, 25% of the students indicated that it is good and a total of 22% indicated that it is excellent.

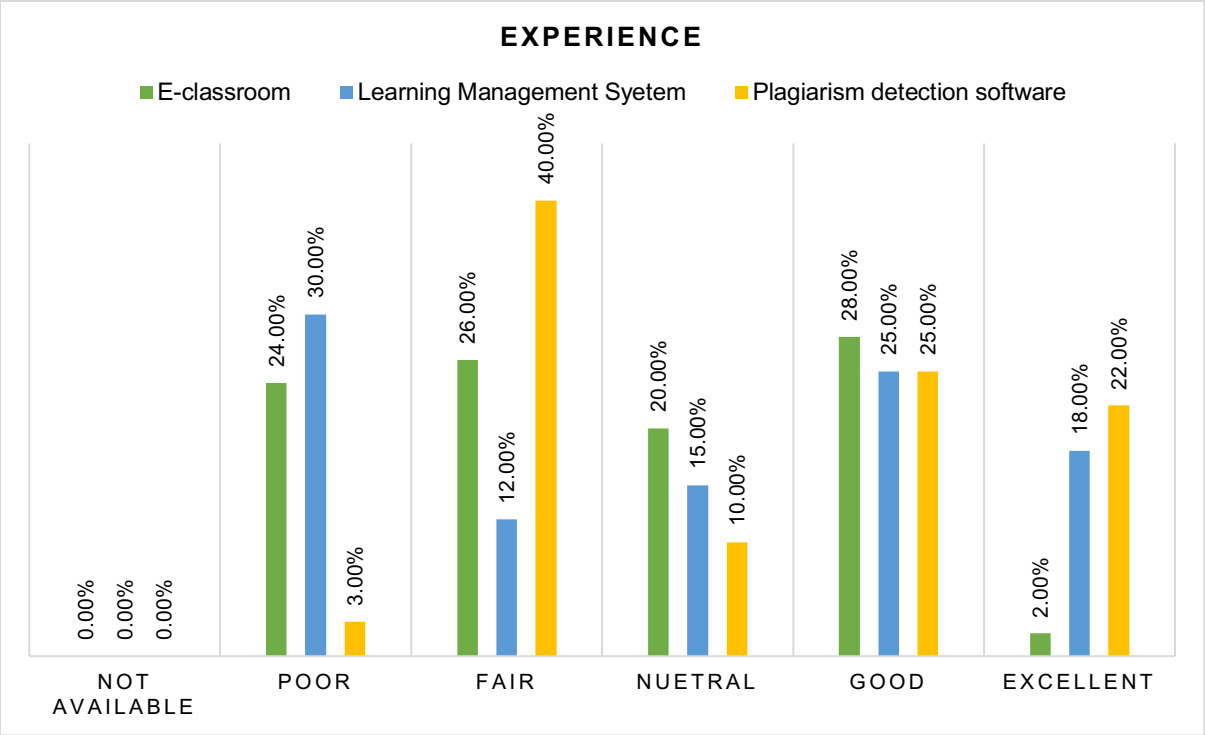


Figure 5. Experience with learning facilities for PIHE

4.3.3 Combined results of learning environment factors

4.3.3.1 Teaching style

Based on the combined findings from IIE MSA and PIHE, a total of 71% of the students indicated that the teaching mode is mixed, a total of 26% indicated that their lessons are taught using the traditional face-to-face and only 3% indicated that their classes are completely online.

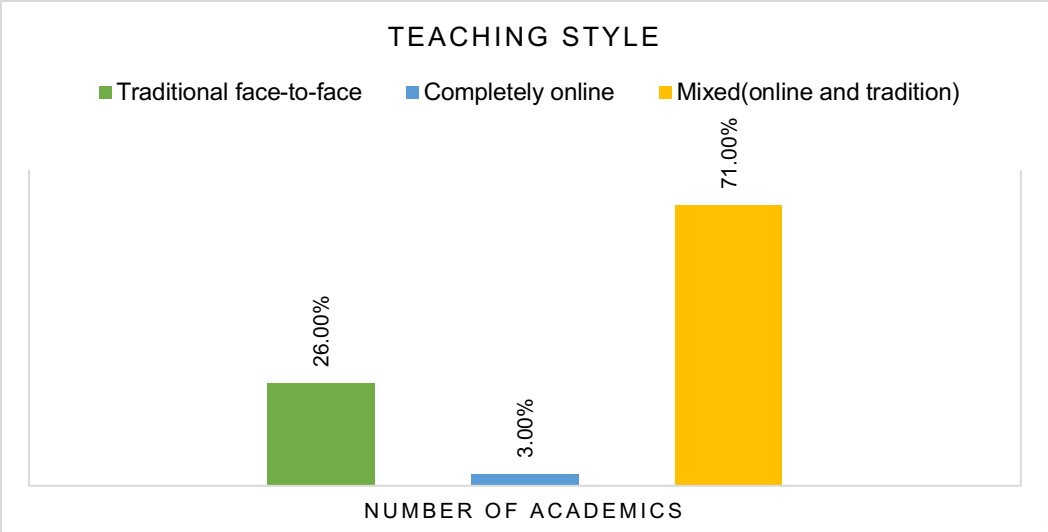


Figure 6. Overall teaching styles

Based on the data collected, it can be concluded that both IIE MSA and PIHE conduct majority of their lessons using mixed teaching styles and only few classes are taught using the traditional face-to-face teaching style.

4.3.3.2 Experience with learning facilities

4.3.3.2.1 E-Classroom

Looking at the data obtained, both institutions provides e-classrooms, 58% of the students from both institutes indicated that the resource was not utilisable, 40% indicated that it was utilisable but not in its best condition, 28% indicated that it was neutral and 56% of the students indicated that the facility is good. Finally, 18% indicated that it is excellent. From the data, it is clear that students from both of the institutes highly utilise these resources during learning process.

4.3.3.2.2 Learning Management System (LMS)

The researcher wanted to determine the level of usage of LMS in the process of teaching and learning. Overall, 68% of the students indicated that the resource is not utilisable. 24% indicated of the students indicated that LMS is utilisable but not in its best condition, 23% indicated that it is neutral and 51% indicated that it is good. The remailing 34% of

the students indicated that the facility is excellent. The data shows that both PIHE and IIE MSA use Learning Management Systems in their teaching and learning process.

4.3.3.2.3 Plagiarism detection software

This question was asked to establish the level of usage of plagiarism detection software in IIE MSA and PIHE. From the data, 5% of the students indicated that the resource cannot be used. Also, 86% indicated that it is utilisable but the facility is not in its best condition, 14% indicated that it is neutral and 53% of the students indicated that it is good. Finally, 42% indicated that the facility is excellent. Clearly both institutions provide very good plagiarism detection facility and the usage level of the facility in the learning process is very high.

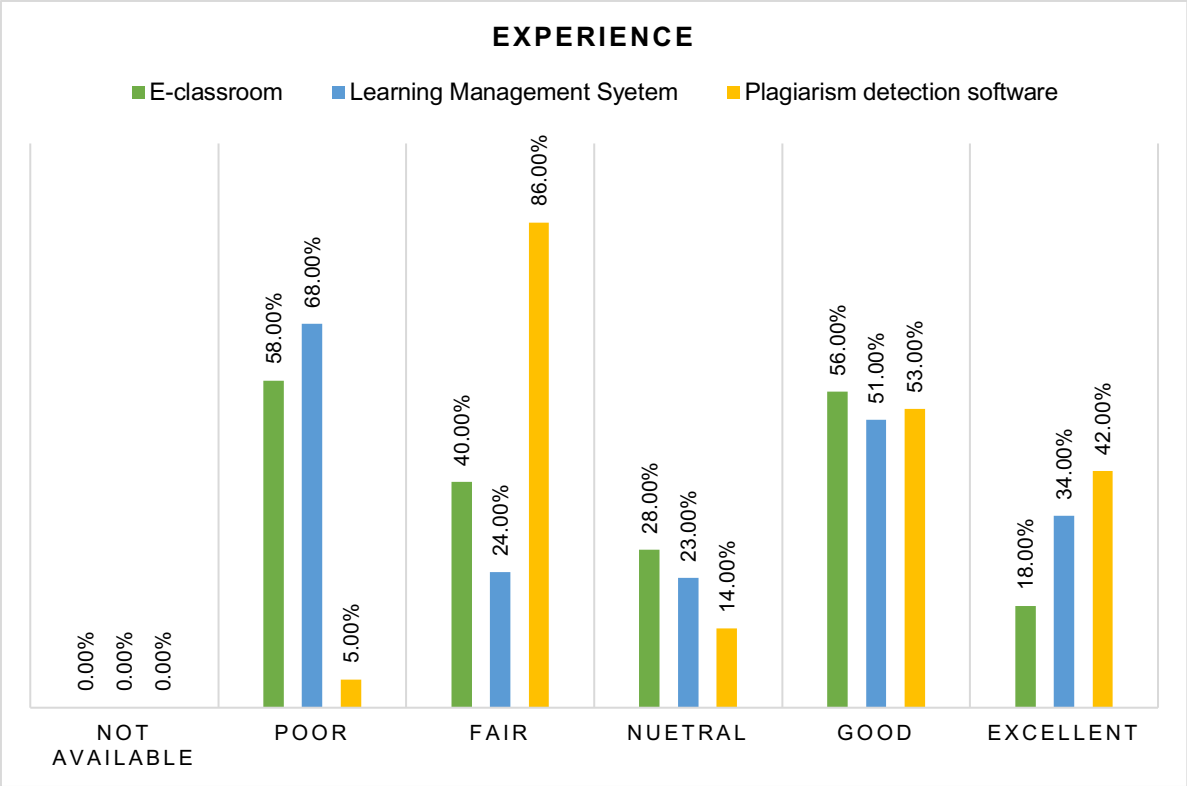


Figure 7. Overall experience with learning facilities

The overall data shows that both these institutions provide good facilities to aid the learning process. But in some case, these facilities are not in good condition. Having good learning facility can impact the learning process by improving students' performance (Ghavifekr and Rosdy, 2015).

4.4 Individual factors

Individual factors show the individual characteristics of the students from the two institutions. This question was important because it will help to explore some of the learning challenges that students face. Also, by exploring individual learning style, these learning styles could be implemented in LMS, giving it the ability to provide a customised learning experience to student. The sub themes under this section include:

- Learning challenges
- Learning styles

4.4.1 Individual factors of students at IIE Monash South Africa

4.4.1.1 Learning challenges

This question was asked in order to find out some of the challenges that students face. From the data collected, a lot of students face challenges with time management (60%), burden of assignments (58%) and coping with reading materials. A total of 34% of the students indicated that they have challenges in becoming an active learner, 28% face cognitive challenges and finally 32% of them have problems with following instructions.

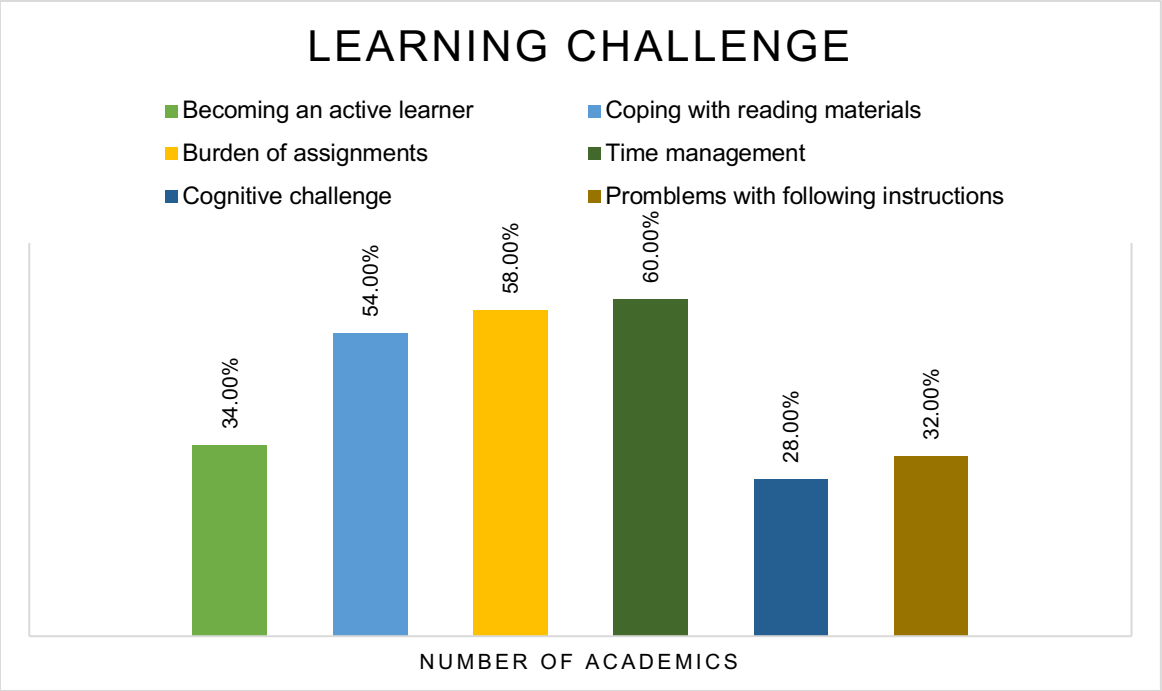


Figure 8. Learning challenges at IIE MSA

4.4.1.2 Learning styles

This question was asked in order to explore the different learning styles of the students. From the data, most of the students are active leaners (26%) and verbal learners (26%). Also, 12% of them are reflective learners, 6% are intuitive leaners, 10% are visual learners, 14% are sensing learners and 6% are sequential learners.

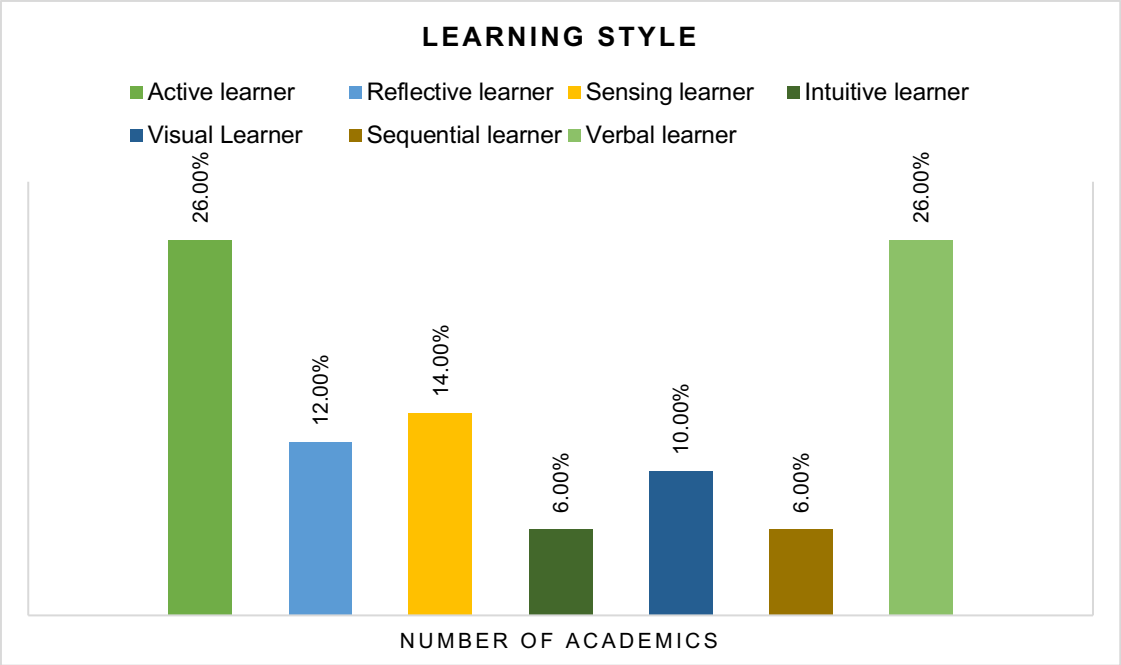


Figure 9. Learning styles at IIE MSA

4.4.2 Individual factors of students at Pearson Institute of Higher Education

4.4.2.1 Learning challenges

From PIHE, a lot of student face challenges with time management (70%), burden of assignments (56%). Also, 48% of students face burden of assignments, 46% have challenges in becoming an active leaner, 44% face cognitive challenges and finally 48% of them have problems with following instructions.

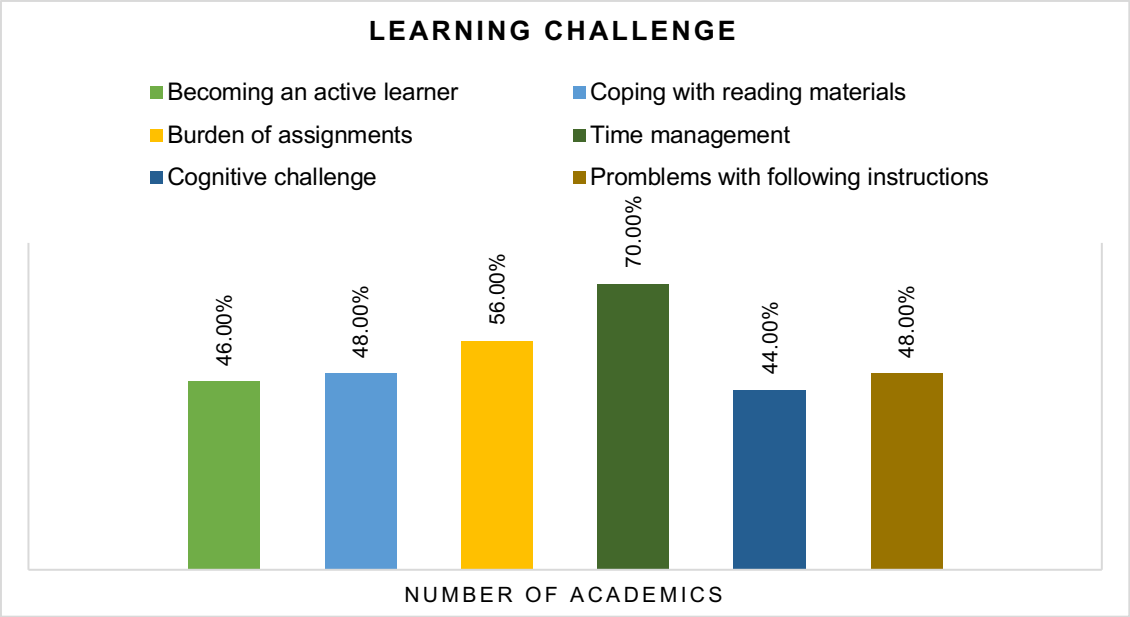


Figure 10. Learning challenges at PIHE

4.4.2.2 Learning styles

The data collected at PIHE shows that most of the students are active leaners (30%) and reflective learners (32%). 8% of the students comprise of sensing learner and 16% of them are verbal learners 16%. Also, 3% are intuitive leaners, 6% are visual learners and 5% are sequential learners.

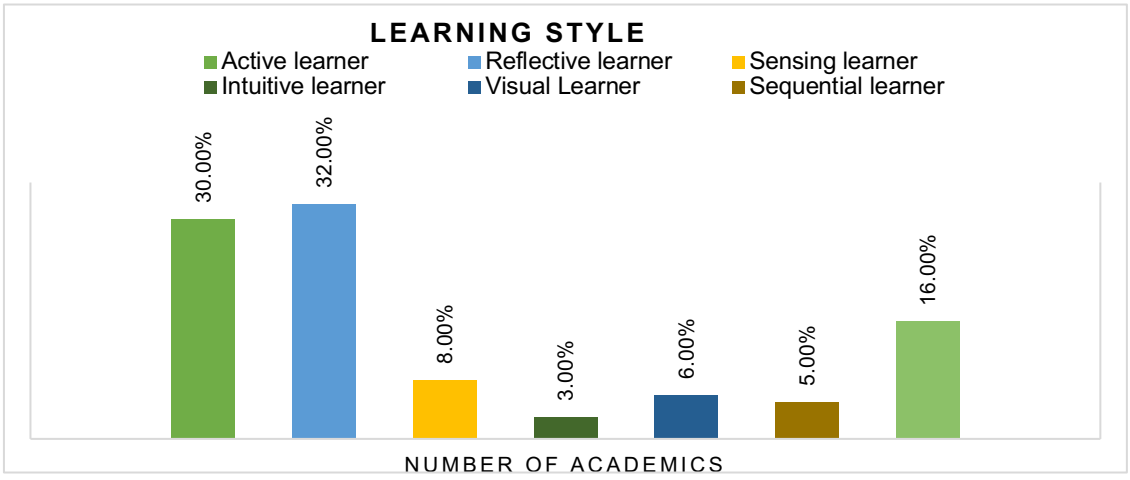


Figure 11. Learning styles at PIHE

4.4.3 Combined results of individual factors of students

4.4.3.1 Learning challenges

Overall, a lot of students face changes with time management (65%), burden of assignment (57%) and coping with reading materials (51%). 40% of them also have challenges with becoming active learners, 36% with cognitive challenges and 40% face problems with following instructions.

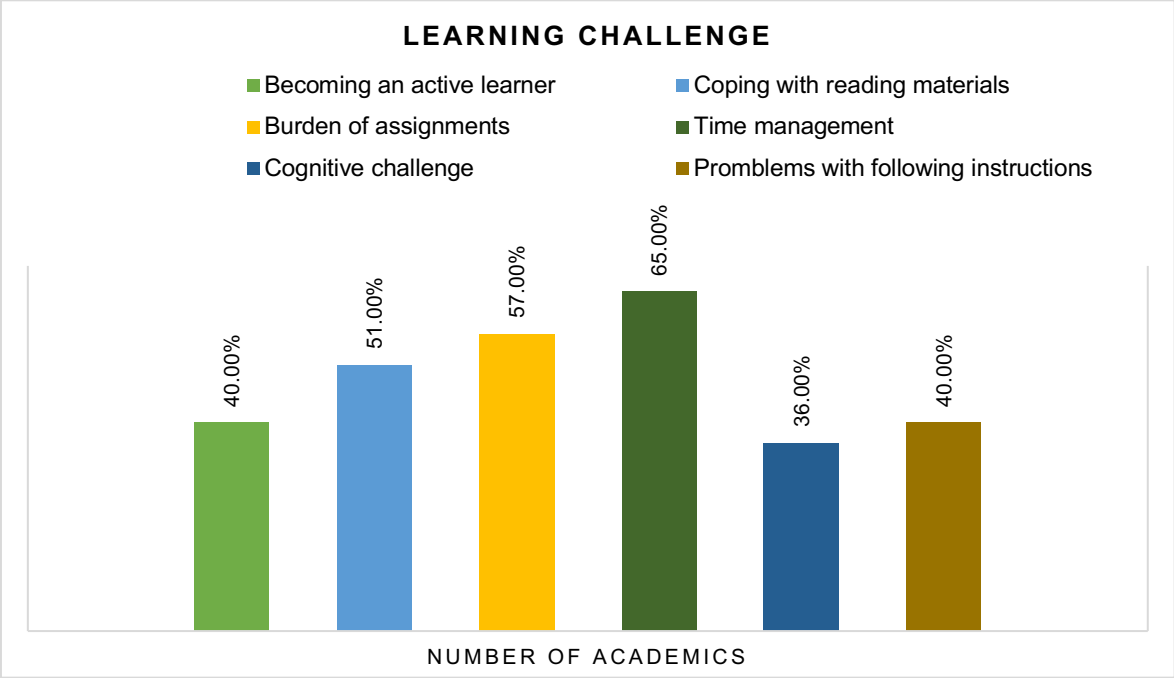


Figure 12. Overall learning challenges

The findings are in line with the results of a study conducted in United States of America on selected University institutions (Fook and Sidhu, 2015). These similar challenges faced by students were also identified in this study. There seems to be a lot of students having problems with time management and experience burn out from assignments.

4.4.3.2 Learning styles

The overall data collected shows that most of the students are active leaners (28%), reflective learners (22%) and verbal learners (21%). Also, 11% of the students comprise of sensing learner, 4.5% are intuitive leaners, 8% are visual learners and 5.5% are sequential learners.

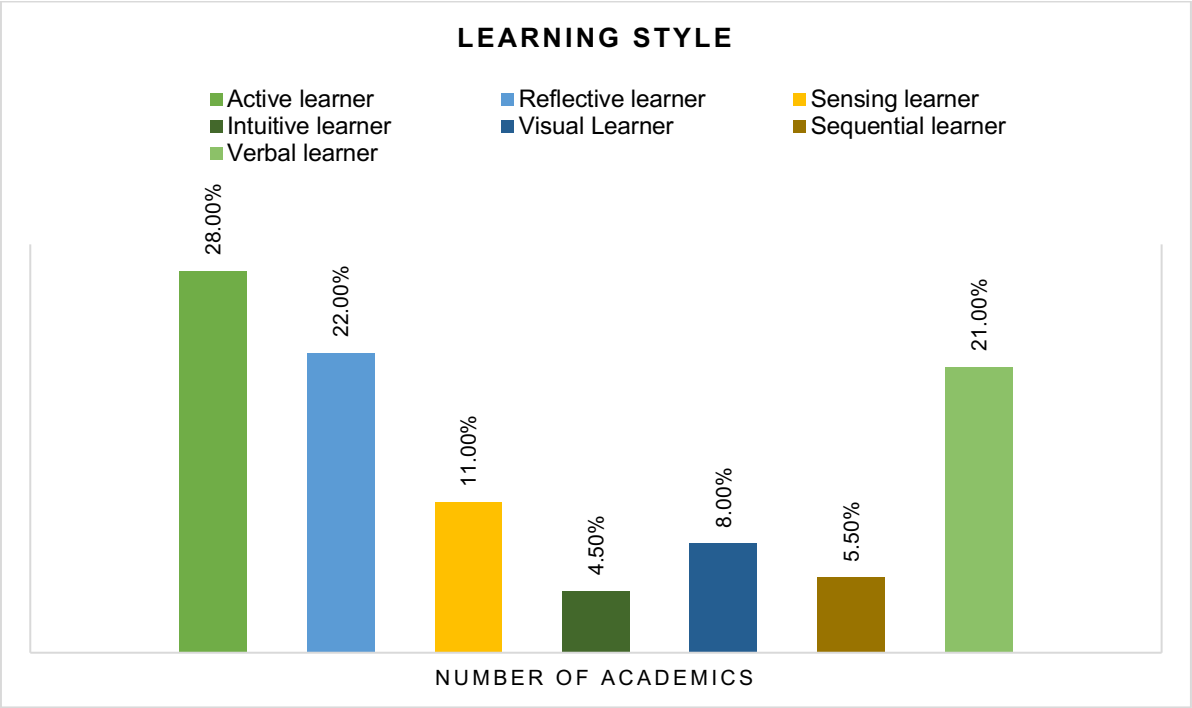


Figure 13. Overall learning styles

These finding shows different types of learning styles that could be used to make LMS more adaptive to suite each individual. These findings agree with the research that was done by Ocepek, Bosnić, Nančovska Šerbec and Rugelj (2013) which showed different learning styles possessed by students. This means that it is important to consider implementing these learning styles in the process of teaching and learning in order to improve performance of the students. This is in line with a study done by Fransoni and Assar in 2009 which explained how it is important to take into consideration different learning styles of students in developing learning materials.

4.5 Individual perception towards the use of technology

This question was asked so that the individual perception towards the use of technology in the process of learning was explored. Different statements on the use of technology was posed to the students to check if they agree with the statements. This would help to understand if individual factors affect the adoption of adaptive LMS.

4.5.1 Individual perception towards the use of technology at IIE Monash South Africa

4.5.1.1 Getting more actively involved in courses that use technology

Students were asked if using technology helps them to get more actively involved with the learning materials. From the data collected, it was revealed that 70% of the students agree with the statement and 8% disagreed with the statement. The remaining students, 20% of them were neutral while 2% of them strongly agreed.

4.5.1.2 Improvement in content engagement through the use of tablets/laptop

The researcher wanted to know if the use of tablet/laptop improves students' engagement with study content. From the data obtained, none of the students strongly agreed with the statement. The majority of them (80%) agreed with the statement while the remaining small portion of the students (20%) were neutral.

4.5.1.3 I wish my lecturers in the university would use and integrate more technology in their teaching

The purpose of this question was to determine if students prefer lecturers to use technology in their teaching. According to the data collected, 60% of students agreed with the statement, while 30% of them strongly agreed with the statement, 4% of the students were neutral while only 6% of them disagreed with the statement.

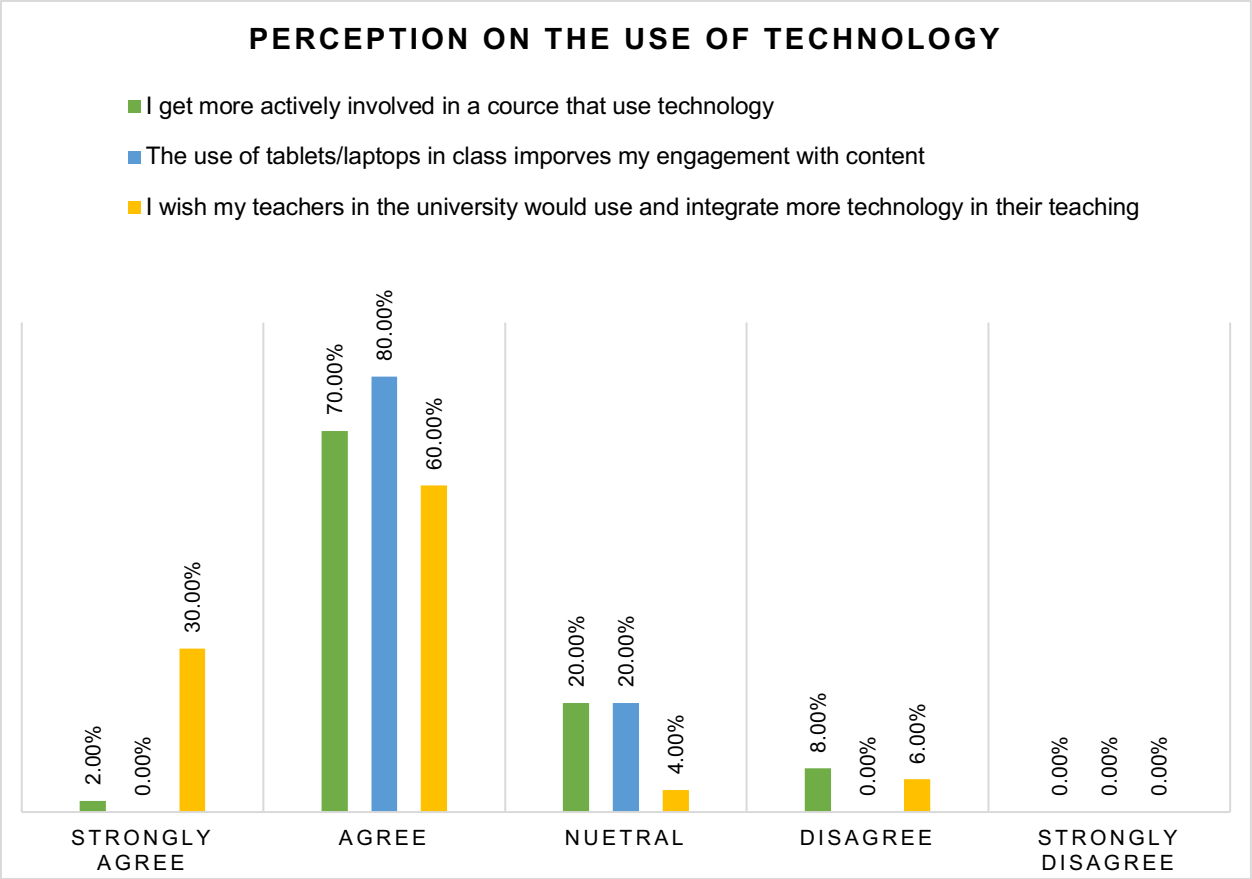


Figure 14. Perception on the use of technology at IIE MSA

4.5.2 Individual perception towards the use of technology at PIHE

4.5.2.1 Getting more actively involved in courses that use technology

When students at PIHE were asked if they get more actively involved in a course that use technology, a total of 10% of them strongly agreed with the statement. A huge portion of students (50%) indicated that they agree with the statement, 30% of the students were neutral and the 10% of them disagreed with the statement.

4.5.2.2 Improvement in content engagement through the use of tablets/laptop

The researcher wanted to know if the use of tablets/laptop improves the engagement of students with their study content. From the results obtained, 90% of the students agreed with this statement while the remaining 10% of the students where neutral. This clearly shows that more students prefer to use technology in the process of learning.

4.5.2.3 I wish my lecturers in the university would use and integrate more technology in their teaching

This question was asked to determine if students prefer their lecturers to use more technology in the process of teaching. Half of the students (50%) strongly agreed with the statement and 30% of students agreed with the statement. The remaining students, 10% of them disagreed and the rest 10% indicated that they were neutral.

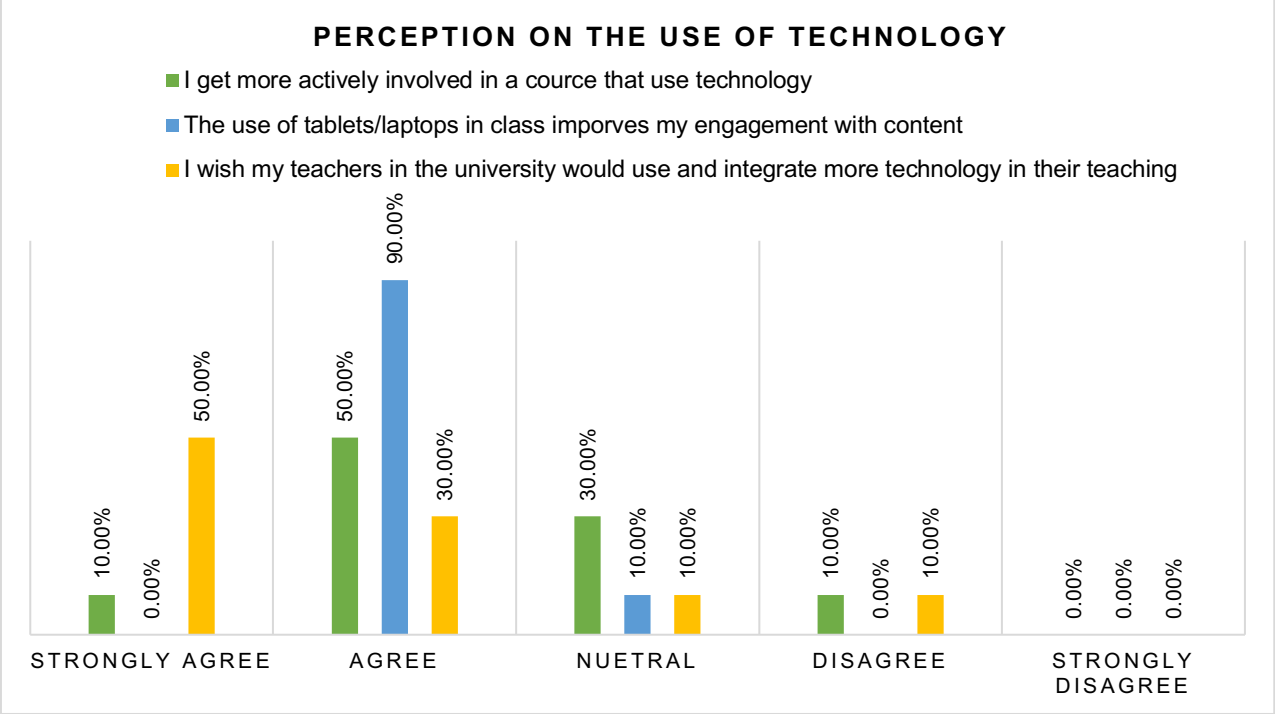


Figure 15. Perception on the use of technology at PIHE

4.5.3 Combined results of Individual perception towards the use of technology of students

4.5.3.1 Getting more actively involved in courses that use technology

Overall, 6% of the students strongly agreed with the statement, and 60% of them agreed with the statement, 25% of them indicated that they were neutral while 9% of them disagreed with the statement.

4.5.3.2 Improvement in content engagement through the use of tablets/laptop

A total of 85% of the students agreed with the statement and the rest of the students (15%) indicated that they were neutral.

4.5.3.3 I wish my teachers in the university would use and integrate more technology in their teaching

A total of 40% of the students strongly agreed with the statement and also 45% of them agreed with the statement, 8% of them disagreed with the statement while the remaining 7% indicated that they were neutral.

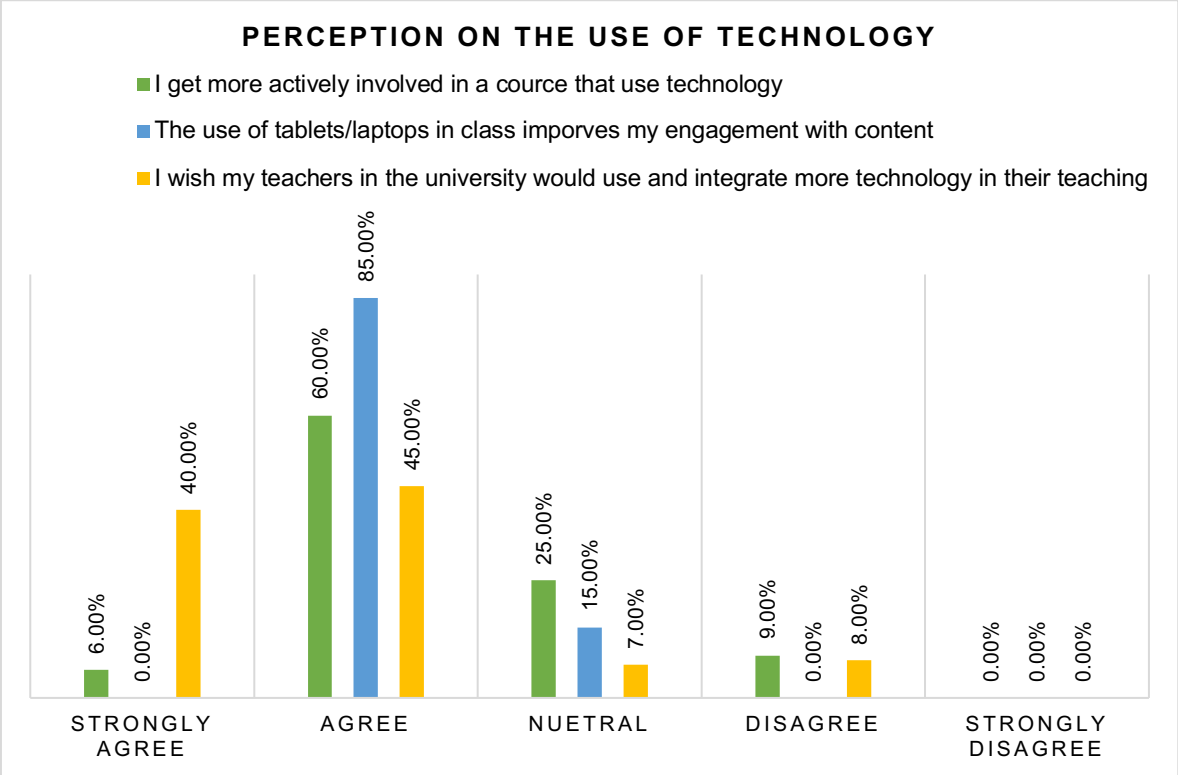


Figure 16. Overall perception on the use of technology

From the data collected, it can be concluded that there is a strong response in the number of students agreeing with the use of technology in higher education. Since the P-value (0.5113) is more than the significance level (0.05), it is clear that data collected is indeed

consistent with this distribution. More details of this distribution are shown in the Table 1 below.

Table 1. Overall perception on the use of technology

Chi Square Results for Goodness of Fit				
H_0: The data is consistent with the specified distribution.				
H_1: The data is not consistent with the specified distribution.				
Alpha	0.05			
c^2	1.342			
$c^2_{(0.05, 2)}$	5.991			
p Value	0.5113			
The null hypothesis is accepted.				
There is no evidence that the data is not consistent with the specified distribution.				

4.6 Usefulness perception of Adaptive LMS

The usefulness perception of adaptive LMS was collected in order to find out if students perceive adaptive LMS to be of any advantage. Secondly, the researcher wanted to establish the way in which adaptive LMS could improve student's academic performance.

4.6.1 Usefulness perception of Adaptive LMS at IIE Monash South Africa

This section shows the nature of the data collected at IIE MSA on the perceived usefulness of adaptive LMS by the students.

4.6.1.1 Customised learning experience

The students were asked if they perceive customised learning experience as being important. The researcher wanted to determine the usage of adaptive LMS in order to assess the type of functions the students might want the system to provide. From the data collected, 72% of students indicated that they see this function useful. 10% agreeing that

the function would be useful to a limited extent, 17% were neutral, 1% indicated that it is not useful and no of the students indicated that it would be very useful to have this function.

4.6.1.2 Receive most suitable sequence of study materials

A large number of the respondent stated that it would be useful to receive most suitable sequences of study materials and this was comprised of 70% of the students, 20% indicated that it would be useful to a limited extent and 10% of the students were neutral. There were no students who indicated that it is very useful or it is not useful.

4.6.1.3 Customised feedback and help

Based on the literature, if students can receive customised feedback and help, this can improve their performance. This statement was asked to determine how useful this function is perceived by the students. A total of 80% of students indicated that it would be useful, 5% indicated that it would be useful to a limited extend and 15% were neutral. None of the students indicated that it is very useful or not useful.

4.6.1.4 Customised solutions to problems

The data collect at IIE MSA indicated that 50% of students viewed receiving customised solution useful to a limited extent. 30% indicated that it is useful and 20% were neutral. Also, none of the students indicated that it is very useful or not useful.

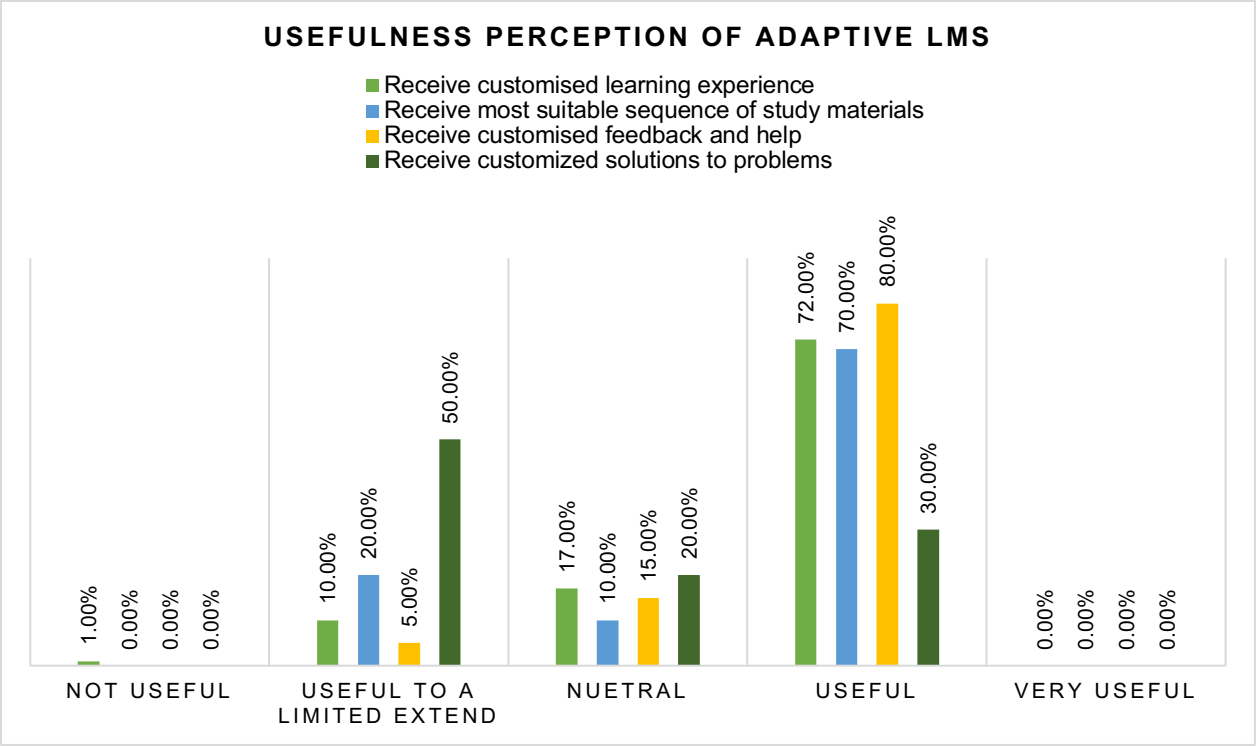


Figure 17. Usefulness perception of Adaptive LMS at IIE MSA

4.6.2 Usefulness Perception of Adaptive LMS at Pearson Institute of Higher Education

This section explains the data that was collected at PIHE on the usefulness perception of adaptive LMS.

4.6.2.1 Customised learning experience

A total of 60% of students indicated that they perceived receiving customised learning experience useful. A total of 10% perceived it as neutral while 30% perceived it as useful to a limited extend. None of the students indicated that receiving customised learning experience is very useful or not useful.

4.6.2.2 Receive most suitable sequence of study materials

A total of 75% of the students perceived receiving most suitable sequence of study material useful. A total of 15% responded as neutral and 10% responded as useful to a

limited extent. Also, neither of the students perceived this as a very useful function and not useful function.

4.6.2.3 Customised feedback and help

From the data collected at PIHE, it indicated that 80% of the students perceived receiving customised feedback and help as useful. A total of 15% viewed it as useful to a limited extent, 5% perceived it as neutral. None of them perceived it as not useful or very useful.

4.6.2.4 Customised solutions to problems

A large percentage of the students (83%) indicated that they perceive receiving customised solutions to problems as useful. A total of 3% indicated were neutral while 14% indicated that it is useful to a limited extent.

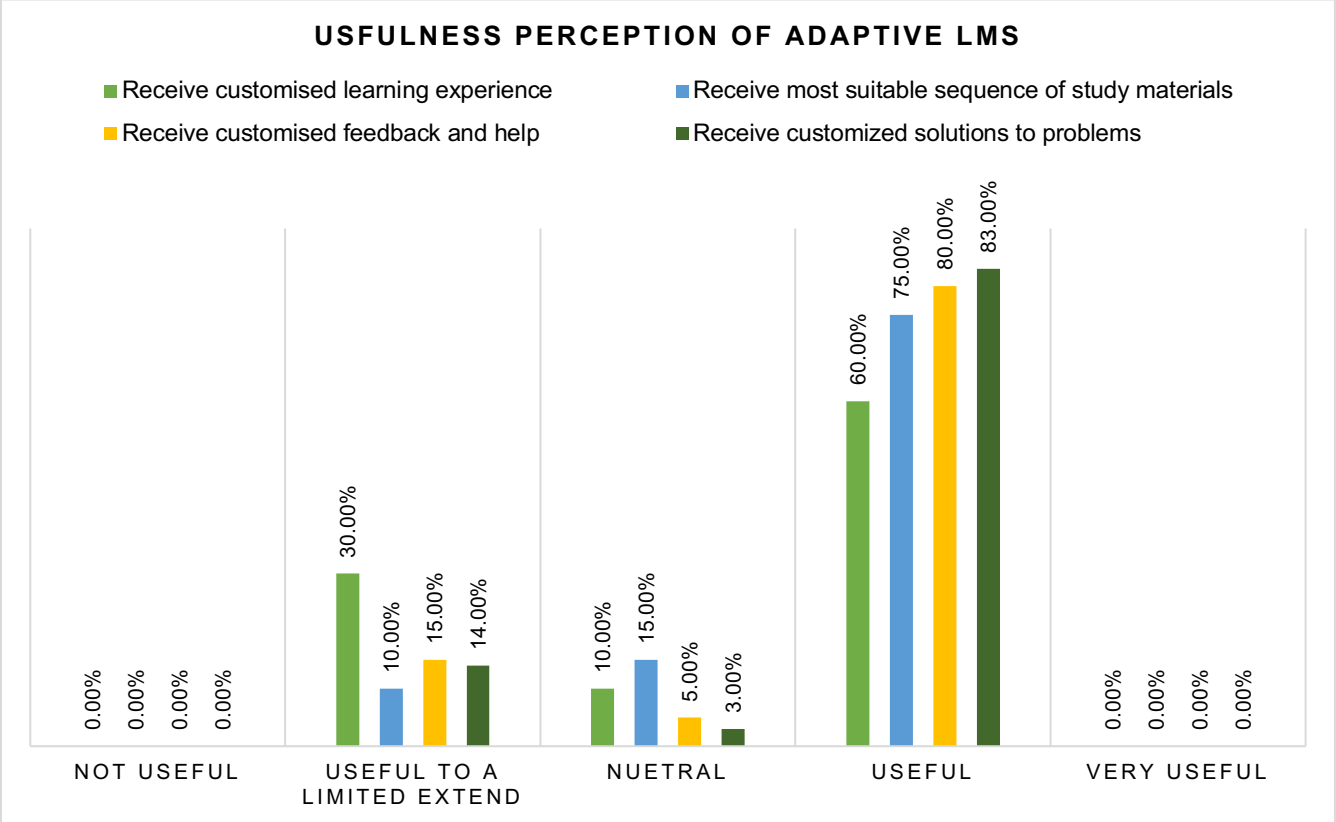


Figure 18. Usefulness perception of Adaptive LMS at PIHE

4.6.3 Combined results of Usefulness Perception of Adaptive LMS of students

4.6.3.1 Customised learning experience

Overall, 66% students indicated that it is useful, 20% perceived it as useful to a limited extent and the remaining 14% indicated that they were neutral.

The combined data also show that there is p value of 0.9168 at a confident level of 95.28 which means that PIHE and IIE MSA are similar in nature and that they are not so different.

4.6.3.2 Receive most suitable sequence of study materials

A total of 73% of the respondent indicated that they preceding this as a useful function and 15% perceived it as useful to a limited extent. The rest of the students (13%) indicated that they were neutral. The combined data also show that there is p value of 0.9168 at a confident level of 95.28 which means that PIHE and IIE MSA are similar in nature and that they are not so different.

4.6.3.3 Customised feedback and help

From the data collected, it revealed that 80% of the students viewed receiving customised feedback and help as useful and 10% of them view it as neutral and useful to a limited extent. The combined data has a p value of 0.9168 at a confident level of 95.28 which clearly indicates that the data collected from PIHE and IIE MSA is similar in nature.

4.6.3.4 Customised solutions to problems

A total of 57% of the students perceived receiving customised solutions to problems as useful, 12% were neural and 32% perceived it as useful to a limited extent. From the p value of 0.8345 at a confident level of 95.28, we can see that the results from PIHE and IIE MSA are consistent under this distribution.

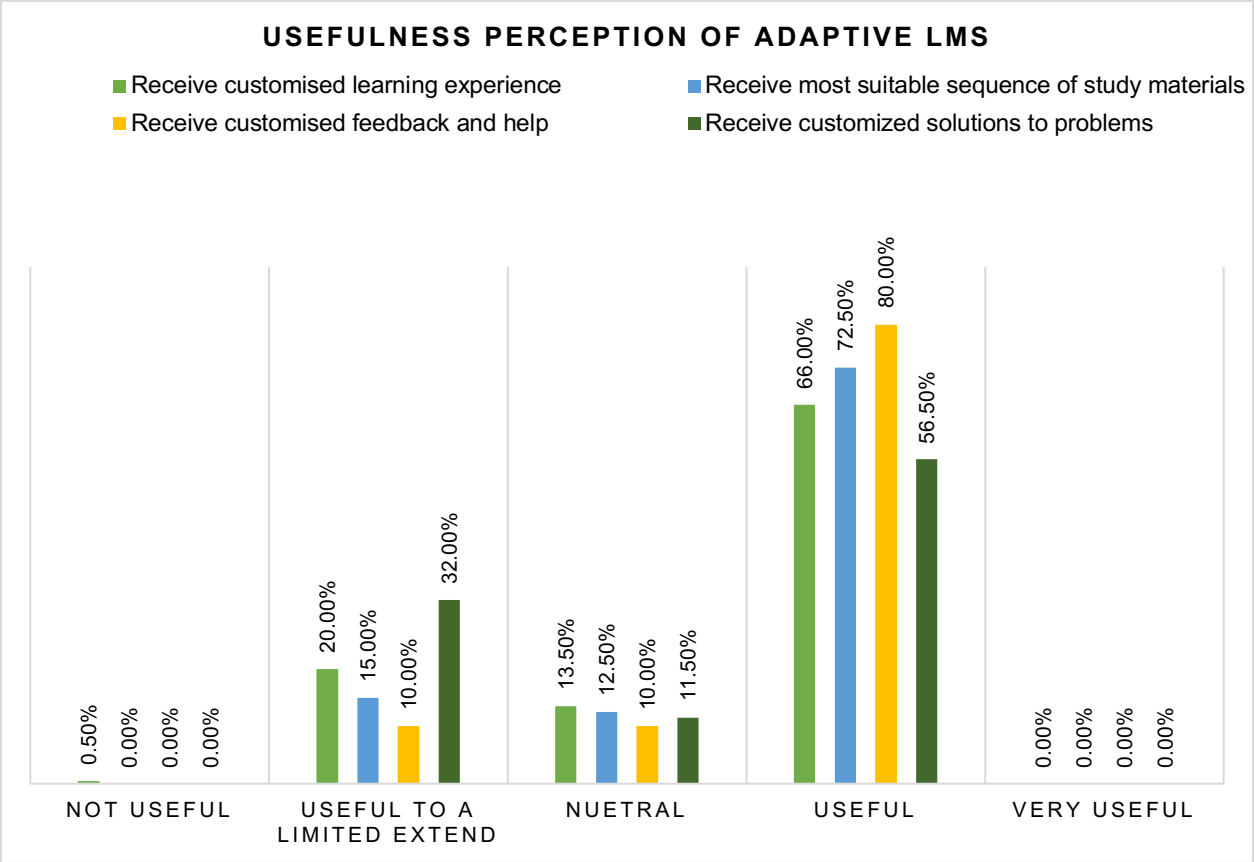


Figure 19. Overall Usefulness perception of Adaptive

There is clearly evidence that students perceive the use of adaptive LMS as being useful which clearly suggest the importance of this system. From Table 4.2 below, it can be seen that the p value is 0.0136 which is less than the alpha value of 0.05. this means that there is evidence of a statistically significant difference in the continuous outcome variable between the five liker scale variables. Thus, students indeed perceive adaptive LMS as useful rather than not useful.

Table 2. Overall usefulness perception of Adaptive

Kruskal-Wallis Test with Pairwise Comparison					
	Not Useful	**Useful to a limited extent**	**Neutral**	**Useful**	**Very useful**
Median	**0**	**17.5**	**12**	**69**	**0**
Rank Sum	8	54.5	45.5	74	0
Count	4	4	4	4	4
Average Rank	2.000	13.625	11.375	18.500	0.000
Alpha	0.05				
kw Statistic	12.575				
p value	0.0136				
Conclusion					
There is evidence that there are differences in the methods.					

4.7 Summary

From the results of the data that was collected from two academic institution, it is clear that the students perceive using an adaptive learning management system as very important as it can help them with improving their academic performance. Also, the students view the use of technology as very important.

The results also confirmed that there are different learning styles and it is important to take them into consideration when developing learning material for students. From the data analysis, we can clearly see that the learning facilities that are provide at these different institutions are not in the best condition and this could be the reason why there has been a delay in implementing adaptive systems. The next chapter will discuss the conclusions and recommendations of the study.

Chapter 5: Conclusions and recommendations

5.1 Conclusion

After the researcher collected and analysed the data, it is evident that adaptive learning management system can play a major role in improving the students' academic performance. From the data collected, it is clear that an adaptive learning technology that can provide customised experiences to students is very important. This answers the research question which looks at the type of adaptive technologies that can be used in higher education.

In order for adaptive learning management system to be implemented, there is a need for good learning facilities to be provided by the institution. From the data we can see that most of the facilities that are available are not in their best conditions and this could be the reason why there has not been much effort in implementing adaptive functionalities in these learning management systems.

5.2 Recommendations

The results obtained from the data collected, it is clear that adaptive learning management systems are very important and should be used in order to improve students' performance. To ensure that the adaptive learning management systems are implemented, institutions should:

- Provide good learning facilities that can promote the use of the adaptive learning management systems.
- Take into consideration the different learning preferences of students during development of study materials, rather than creating general study materials for all students
- Provide incentives to lecturer to see the need to use more technology in their teaching process.

Appendices

Appendix A: Ethical Clearance from Research Committee

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14 October 2019

Pearson Institute of Higher Education
Faculty of Applied Science
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Student Name: John Paulo
Student Number: G21RL3WP4
Title: Factors that influence the adoption of Adaptive Learning Systems in higher education Degree: BSc IT (Honours)
Module: ITRP420 (Research project)

FASREC reference number: IT/08/2019

Ethics Approval

We are pleased to inform you that the application for ethical clearance for the above- mentioned protocol is approved for a period of 1 year. Should there be any amendments to the protocol, FASREC needs to be notified prior to the changes being implemented. FASREC strives to comply with the South African Research Guidelines (2015) and the FASREC terms of reference.

We hope that the study goes according to plan and wish you everything of

the best. Yours Sincerely



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Appendix B: Draft questionnaire

Title: Questionnaire on Learner Use of Learning Management System/Technology

Introduction: An algorithm can be defined as a set of rules or a process that are performed using a computer in calculations or in problem-solving operations. Adaptive learning is a type of learning that involves the use of algorithms and computers to facilitate and deliver customized learning materials based on student's preferences. When the machine learning algorithms are integrated in Learning Management Systems (LMS), they can make it easier for teachers to provide materials to students in a customized way, making it easier for students to understand these academic materials.

Aim: The primary aim of this questionnaire is to assess the Technology-Enabled Learning environment and preferences for adopting technologies for learning in an educational institution. The questionnaire shall be completed by a randomly selected stratified sample of learners in two institutions IIE MSA and Pearson institute of higher education.

Protecting the privacy of the respondent is important — all personal information will be kept confidential.

Please respond to all the questions by following the instructions.

A. Background information

1.1 Name of the University/institution ◯ IIE MSA ◯ Pearson Institute of higher education

1.2 Most of the courses you are currently studying are: **Select only one that applies to you**

◯ Traditional face-to-face

◯ Completely online

- ○ Blended, where some components of the study are done online

1.3 What are the earning challenges that you are currently facing in your study? **Select all that applies**

- ○ becoming an active learner
- ○ coping with reading materials
- ○ following instructional problem
- ○ burden of assignments
- ○ cognitive challenge
- ○ time management

1.4 What Is your learning style? **Select the one that appropriately describes your learning style**

Place an "X" in the box of your answer

Learning style	Description of learning style	x
Active learners	These leaners understand and retain information well through a practical interaction with the learning material	
Reflective learners	The leaners prefer to think and process the learning material quietly in form of reflection	
Sensing learner	These leaners prefer real-world facts so that they can apply the learning material in a more practical sense	
Intuitive learners	They prefer material that do not require routine calculations and memorization	
The visual learners	Prefer the use of diagrams, pictures, and charts in order to understand the learning materials well. They like the use of visual representations in leaning	
The verbal learners	They understand better when the learning materials are in the words format; both spoken and written	
Sequential learners	They prefer learning in a linear way, going one chapter to another logically	
Global learners	The leaners prefer learning in a randomized way. Jumping from one topic to another, without following the normal topic arrangement	

B. Access to and Use of Information and Communication Technologies (ICTs)

Technology-Enabled Learning Environment

1. Please rate your experiences with the following resources/services/spaces provided by your institution.

Resources/Services/Spaces	Not Available	Poor	Fair	Neutral	Good	Excellent
eClassroom facilities (e.g. SMART boards, projection systems, computers, lecture capture systems etc.)	☐	☐	☐	☐	☐	☐
Learning Management System (e.g. Moodle, myLMS, etc.)	☐	☐	☐	☐	☐	☐
Plagiarism detection software	☐	☐	☐	☐	☐	☐

C. Perceptions of Use of Adaptive Learning Management System

1. Please indicate to what extent you agree with the following statements.

Statements	Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree
I get more actively involved in courses that use technology.	☐	☐	☐	☐	☐
Use of tablets/laptops in class improves my engagement with the content and class	☐	☐	☐	☐	☐
I wish my teachers in the university would use and integrate more technology in their teaching.	☐	☐	☐	☐	☐

2. Please rate how useful each of the following technologies currently is or would be in your studies (regardless of whether or not you have used each technology in the past)

In your studies, how useful do you think it would be to …	Not at all useful	Useful to a limited extent	Neutral	Useful	Very useful	Do not know
Receive a customized learning experience?	☐	☐	☐	☐	☐	☐
Receive most suitable sequence of the study materials/Curriculum that suits your learning style?	☐	☐	☐	☐	☐	☐

Receive a customized feedback and further help from LMS on different assessments?	☐	☐	☐	☐	☐	☐
Receive customized examples to solving a problem depending on level of your understanding of the problem at hand?	☐	☐	☐	☐	☐	☐

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